CISCO WEBEX · CASE COMPETITION 2025

VIBE

VOICE INTELLIGENCE & BOOST ENGINE

No more **"am I audible?"** — VIBE is the real-time AI voice & language layer that cleans, translates, simplifies, and speaks for you.

A · Setup & Entry | B · Live Audio Path | C · Type-to-Speak | D · Q&A Loop | E · End & Logging | AI Processing | Decision Gate

⬡ START — VIBE LAYER INITIALIZED

◈ PHASE A · SETUP / USER ENTRY

1 🖥️ **User Joins Webex Meeting**

VIBE layer auto-detects audio environment · System profiles mic + room acoustics

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2 🎛️ **User Toggles VIBE Layer & Selects Mode**

Choose one or more active modes:

🔊 Voice Boost | 🌐 Live Translate | ⌨️ Type-to-Speak

🧩 Simple Mode (Non-Tech) | 🤹 AI Avatar / Double

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3 🌍 **User Sets Language & Profile**

Native Language: Hindi · Spanish · Mandarin · French · Arabic · + 40 more

Tech Level: Technical ↔ Non-Technical

Voice Preset: 🎙️ Podcast Voice | ✨ Crystal Clear | 😎 Chill Voice | 🏢 Pro Mode

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❓ **Is user speaking or typing?**

🎙️ SPEAKING → Path B: Live Audio

⌨️ TYPING → Path C: Type-to-Speak

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◈ PHASE B · LIVE AUDIO PATH · WHEN USER SPEAKS

1 🎤 **Microphone Audio Captured**

Raw PCM audio stream · Per-speaker separation (Webex native) · Real-time buffer

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2 🤖 **Audio Intelligence Front-End**

- ✦ Noise Removal (café / wind / keyboard / crowd)
- ✦ Echo Cancellation + Auto Gain Control
- ✦ Background Blur: Music · HVAC · Dog · Baby
- ✦ "Optimize for My Voice" — learns your vocal signature

NEW → Smart Room Detection: detects reverb, adjusts EQ live

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3 🔊 **Voice Boost & EQ Engine**

Apply selected preset → Loudness · Clarity · Warmth · Presence

Adaptive EQ: noisy café ≠ quiet office ≠ conference room

Real-time waveform normalization · Consistent output level

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4 🗣️ **Speech-to-Text (ASR)**

Real-time transcription in user's native language

Multi-accent support · Noisy input handling

Per-speaker diarization (Who said what?)

NEW → Filler word detection: removes "um", "uh", "like" from transcript

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5 🧠 **NLP & Language Understanding**

Detect: intent · domain · tech level · sentiment · key entities

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🧩 IF SIMPLE MODE ON

🌐 IF TRANSLATE ON

📊 ALWAYS ON

Jargon Simplifier

Rewrites technical text
Adds real-world analogies

Example: "API" → "a waiter between kitchen & table"

NEW→ Emoji + visual cues added

Live Translator

Native lang → English (or any target language)

Context-aware translation
Preserves technical terms

NEW→ Tone preservation

Confidence Booster

Detects hesitation & suggests rephrasing in real-time

Highlights unclear segments

NEW→ Private earpiece coaching mode

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6 🔊 Text-to-Speech Output (VIBE Voice)

Converts processed/translated text → natural speech
Consistent "VIBE Voice" identity per user across sessions
Target: English (global) or any specified target language

NEW → Emotional tone matching — excited text sounds excited

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7 📡 Deliver to Other Participants

They hear: boosted · cleaned · translated VIBE voice
Live captions in their own language (each attendee can set their own)

NEW → Caption language is per-user — everyone sees their own language simultaneously

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◈ PHASE C · TYPE-TO-SPEAK PATH · WHEN USER CANNOT / WON'T SPEAK

1 ⌨️ User Switches to Type-to-Speak Mode

Triggers: muted · too noisy · mic broken · speech difficulty · shy in large call

NEW → Auto-suggest: VIBE detects long silence + background noise → offers this mode

2 📝 **User Types in VIBE Panel or Chat**

Typing in any language · Can paste code snippets, data, tables

3 🧠 **NLP Processing**

Detect language & intent · Simplify or professionalize based on mode

Translate to meeting language (e.g., → English)

NEW → "Polish mode" — convert quick draft into fluent spoken language

4 🤖 **VIBE Agent Speaks on Behalf of User**

Agent voice speaks typed message aloud in the meeting

Screen label: "🎙️ Spoken by VIBE for [UserName]"

User can choose from 3 voice personas for their VIBE agent

✦ NEW FEATURE

⚡ **VIBE Interrupt Signal**

If user types "!" prefix → VIBE flags an urgent interrupt

Host is notified: "[UserName] wants to raise a point"

Smooth, non-disruptive way to speak up in large meetings

5 📋 **Caption & Chat Sync**

Caption displayed: "from [UserName] via VIBE ⌨️"

Auto-appended to meeting transcript with source tag

◈ PHASE D · Q&A LOOP · FOR NON-TECH / NON-NATIVE USERS

1 🙋 User Asks in Own Language (Voice or Text)

Example: "Yeh API integration kaise kaam karti hai?" (Hindi)
Example: "¿Cuánto cuesta el plan enterprise?" (Spanish)
VIBE detects language → routes to translation pipeline

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2 🔄 ASR + Translation to English

Native speech → English text · Displayed to presenter as caption
Presenter sees: "[Priya] asked (translated): How does the API integration work?"

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3 🎤 Presenter Answers in English

Presenter responds normally · VIBE listens and captures the answer
NEW → Presenter can tag: "Simplified answer" to trigger extra NLP treatment

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4 🌐 Translation + Simplification Back to User

English answer → translated to user's native language
If Simple Mode: adds analogy + concrete example
Example: "Think of an API like a power outlet — your device doesn't need to know how electricity is generated"

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5 📤 Output to User

🔊 Audio in native language via VIBE voice
📝 Text with simplified explanation + examples in caption pane
🔗 Optional: "Save this explanation" bookmark

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✦ NEW FEATURE

🤝 VIBE Empathy Score

After each Q&A cycle, VIBE privately scores clarity of the presenter's answer (1–10)
Presenter sees: "Your answer scored 6/10 for clarity to non-technical users."
"Try adding a visual example next time." → Coaching in real-time

◈ PHASE E · END & LOGGING

1 📼 VIBE Transcript Saved

Original language layer
English translation layer
Simplified text layer
Timestamps + speaker tags

2 📊 Session Analytics

Voice clarity score over time
Translation accuracy report
Q&A resolution rate
Empathy score trend

3 📦 VIBE Summary Pack

Key Q&A in native language
Simple explanations + examples
Action items extracted
Share via Webex / email

✦ NEW FEATURE

🔁 VIBE Personal Memory (Cross-Session)

VIBE remembers your voice preset, preferred language, and jargon glossary across all meetings
"Your VIBE profile is active — Crystal Clear voice, Hindi ↔ English, Non-Tech simplification ON"
Enterprise admins can set org-wide VIBE defaults for all employees

✓ END — VIBE SESSION COMPLETE

VIBE positions Webex as the only platform where
language, noise, and tech barriers never limit who can participate.

Cisco Webex · VIBE Feature Proposal · Case Competition 2025